

CHAPTER 1.1

1. Difference between hardware and software
 - Hardware are the physical electronic components that make up the computer
 - Software are the programs that run on the machine allowing it to produce useful results
2. External Storage
 - A computer system would need an external storage to store data even when the power for the system is off
 - Transfer data from one machine to another more easily
3. Difference between system software and application software
 - System Software controls how to computer system works
 - Application Software is what allows the user to carry out tasks on the computer

CHAPTER 1.2

1. Purposes of an Operating System (OS)
 - Creates an environment on the computer in which application software can be executed
 - Manages the physical resources of the computer system
 - Provides a means of communication between the user and the computer
2. Types of OS
 - Batch
 - Does not allow for interaction between user and the processor during execution of the program
 - Inputs lots of data and the Batch OS controls their passage through the computer
 - Like trying to interact with a program that lagged out and is locked out
 - Real-Time
 - Reacts quickly enough to affect the next input or process that needs to be carried out
 - Continuous cycle of input-processing-output
 - Example: Sensor detects light and outputs a sound, a delay in the input-processing-output system will cause the sound to break

- Repeat + Always on
- Single-User
 - Controls a system with 1 user and multiple programs at any one time
 - Laptop at home only you can access
- Multi-User
 - More than one user to access the system at the same time
 - Single computer with multiple terminals that multiple users can access at the same time
 - Google sheets with friends
- Multi-Tasking
 - Gives the user the impression that they can carry out more than one task at the same time
 - Similar to single user, but imagine you can open multiple tabs and run them all at the same time
- Network
 - Multiple computers all linked together for the purposes of communication and the sharing of resources
 - Allow hardware and software to be shared
 - Often cases 1 machine is the server host and the rest join the network

3. Types of User Interfaces (UI)

- Form Based Interface
 - An on screen standardised form to be filled in
 - Commonly used by a telecom operator while talking to customer over the phone
 - Prompts the operator to ask each of the questions in turn
 - Makes operator input in correct order
 - Ensures the right format is used
 - May have specific inputs in order to restrict input to ensure correct format
 - Tick boxes etc.
 - Google forms :)
- Menu Based Interface
 - Used in cases where user does not know what options are available
 - Like category shopping lists on websites
- Graphical UI (GUI)
 - Windows, Icons, Menus, and a pointer
 - Like photoshop or Minecraft GUI
 - Keeps different apps/files in a bordered area called a **window**
 - User selects a command by clicking on an **icon** in the **window** using the **pointer**

- User selects further options in **menus**
- Natural Language Interface (AKA conversational Interface)
 - Displays or speaks questions that gives the user the impression that the computer is talking to them
 - Eg.: Phone companies when they call and it says “Press 1 for english, 2 for chinese etc.”
 - System restricts itself to limited inputs with simple responses/outputs
 - Computer expects very specific language inputs which is difficult because people speak/type differently

4. Utility Software

- Disk Formatter
 - When disks are first produced they have no data and cannot save data until a formatter is used to format the disk
 - A formatter splits the area on the disk into “Tracks” which then can be used to store data
 - 1 Track will be used as an index, it holds data on where the other “Tracks” are
 - Formatting a disk wipes all of its previous data
- Hardware Drivers
 - Control communication between the hardware device and the OS
- File Handling
 - A computer stores its data as “Files”
 - An application like “Finder” or “Explorer” is used to manipulate the data
- Automatic Backup or Archive
 - An application/archiving utility is a simple routine that copies the files and stores them somewhere else
 - If the file is damaged, it will use the backup file to replace the corrupted one
 - Incremental backups, automatically backs up changes made to the files since the previous backup copy was made
- Anti-Virus Software
 - It compares a dictionary of known viruses to all the accessible files on the computer
 - It alerts the user of the files that match the virus’s signature
- File compression software
 - It cuts out majority of the duplication of data in a file

CHAPTER 1.3

1. Representation of Character Sets

- Bits
 - 8 bits = 1 byte
 - 1 byte stores 256
 - Eg. 0001 0101 = 21
- ASCII (American Standard Code for Information Interchange)
 - 8 bits but first bit is used for checking
 - 7 bits = 128 characters
- Unicode
 - 16 bit = 2 bytes
 - Can represent 65,000 characters

2. Representation of different Data Types

- Numeric Data
 - Integers
 - Floats
 - A byte can store 0 to 255
- Boolean Data
 - True or False
 - 1 bit
- Character Data
 - "A", "1", "%"
- String Data
 - A sequence of Characters stored together
- Others
 - Date/Time
 - Currency

3. Representation of Positive Integers

- Denary 0, 1, 2, 3, 4, 5, 6, 7, 8, 9
- Binary 1, 0

4. Structure of an Array

- Made up of an index and a subscript
- List[0] will pull the first subscript (index 0)
- Must Initialize Array before storing data so that the computer knows what to put inside
 - Reason why some python code uses list = []
 - Creates a blank array which will be filled up later

- `Array.append("Josh")` adds the string "Josh" to the back of the array
- Searching in an array
 - Using Python and an array called `Names = ["Lewis", "Colleen", "Haley", "Joshua", "Evan"]` check if the name "Joshua" is inside

```
Names = ["Lewis", "Colleen", "Haley", "Joshua",
        "Evan"]
found = False
For name in Names:
    If name == "Joshua":
        found = True
print(found)
```

- Queues
 - Start point and End point
- Stacks
 - First in Last Out

5. Accessing Data

- Serial Access
 - Data is accessed in a chronological (time-based) order
 - No fixed structure
 - Reading data from a sensor in real-time or processing log files line by line as they are written.
- Sequential Access
 - Data is accessed in a fixed, linear order
 - Must traverse all preceding data to reach a specific record.
 - Efficient for batch processing but slow for arbitrary lookups.
 - Like names in alphabetical order
- Indexed Sequential Access
 - Combines sequential and random access using an index for faster lookups
 - Data is stored sequentially, but an index structure allows direct jumps to specific records.
 - A phone book with names in order but an index for quick lookups.
 - Databases where records are ordered but need faster search
- Random Access
 - Any data item can be accessed directly without reading preceding items
 - Uses addresses
 - No dependency on physical order.
 - Modern databases, file systems, and applications needing fast lookups
 - Json files

CHAPTER 1.4

1. Components of a processor

- Control Unit
 - Manage Instructions
 - Make sure other parts of CPU are functioning properly
 - Decides what instructions to carry out next
 - Decodes instructions to work out what needs to be done to execute them
 - Sends out control signals to other parts of the processor to inform them how to carry out the instructions
 - Synchronises instructions from its internal clock
- Arithmetic and Logic Unit (ALU)
 - Add numbers together
 - Allows for decisions to be made
 - Enables logical comparisons
 - Acts as gateway between processor and other parts of the computer
 - All inputs and outputs goes to ALU and waits for Control Unit to decide what to do
- Main Memory
 - Immediate access store (IAS)
 - Stores the data that the processor is going to use
 - All program instructions and data needed to carry out said instructions
 - OS modules are stored too

2. Types of Primary Memory

- Read Only Memory (ROM)
 - Non-volatile
 - Cannot be changed or altered
 - When computer is turned off ROM is not affected
 - Small program stored in ROM enables the OS on hard drive (Bootstrap Loader)
- Random Access Memory (RAM)
 - Volatile
 - Larger Storage then ROM
 - Stores user files that change
 - User data and files are stored

3. Secondary Storage Media

- Magnetic Tape
 - Data is stored Sequentially
 - Large Volumes of data
 - Good for backups but bad for short lookups
- Magnetic Hard Disk
 - Large amounts of data on a single surface
 - Read and Write heads
 - HDD
- Optical Disc
 - DVD more storage than CD
 - CD/DVD-ROM cannot be altered
 - CD/DVD-ROM is used to distribute large data files like audio books or encyclopedias
 - CD/DVD-RW can be read and rewritten
 - CD/DVD-RW is used for transporting files or backups
 - CD/DVD-R can read and write but cannot be rewritten
- Solid State Technology
 - Thumbdrive
 - SSD
 - Fast
 - Stored on a circuit similar to primary memory
 - Portable and very small
- DVD-RAM Disc
 - Surface can be rewritten far more than DVD-RW
 - Faster
 - Reading and writing can take place simultaneously
- Blu-ray Disc
 - Optical Storage
 - Larger storage
 - High speed data transfers
 - HD videos
 - Diff between Blu-ray and DVD is the laser used

4. Transferring of Data from Primary to Secondary Memory

- The Processor uses primary data so it plans to receive data from secondary to primary
- Primary is faster than secondary so Processor does something else while waiting
- Secondary sends data to a **“Buffer”** in Primary storage
- If all the data can be sent to buffer
 - Processor then uses the data
 - Empty buffer to storage

- If data is too big for buffer
 - Buffer is full but there's more data to be sent
 - Processor uses available data
 - Once data is empty storage sends “**Interrupt Service Routine**” to control unit
 - Control Unit services most important interrupt and fills the buffer with more data
 - Meanwhile Processor processes the data

5. Peripheral Devices

- Input
 - Keyboard
 - Mouse
 - Barcode Scanner
 - Optical Character Reader
 - Automatic Data Capture Method
 - Optical Mark Reader
 - Automatic Data Capture Method
- Output
 - Screen
 - Printer
 - Speaker

6. Speed Mismatch

- If inputs are faster than Processor it can miss some inputs
 - Can be solved by improving/optimising code
 - Can be solved by Buffers
- If inputs are slower than processor it creates a bottleneck

CHAPTER 1.5

1. LAN vs WAN

- Allows computers to communicate with one another
- Share information centrally
- Share copies of software
- Give access to data and program files
- LAN (Local Area Network)
 - Network over short distances (Same building)
 - Can Share hardware
 - Privately Controlled
 - High Speed, Low Latency
 - Scalability Issue
 - Requires infrastructure
- WAN (Wide Area Network)
 - Long distances (Regional)
 - Cannot share hardware
 - Centralised Data Access
 - Slower than LAN

2. Types of Data Transmission

- Serial
 - Sends 1 bit at a time down a single wire/connection
 - Slow but accurate
- Parallel
 - Sends multiple bits at a time down multiple wires/connection
 - If there are 8 wires it can send 8 bits
 - Faster than Serial but bits can arrive at different times due to natural error so the bits can get muddled up
- Modes of transmission
 - Simplex
 - Can only be sent in 1 direction
 - Full-Duplex
 - Data can be sent both ways at the same time
 - Half-Duplex
 - Data can be sent both ways but only 1 at a time

3. Bit rates and time sensitivity

- 1 baud = 1 bit per second (1 bps)
- 1 Kbps = 1,000 bps
- 1 Mbps = 1,000,000 bps
- What affects bit

4. Checking for Data

- Echoing back
 - Sends a copy of the data back and checks if its the same
 - If not the same send data again
- Checksum
 - Adds all the bits in the block of data
- Parity Check
 - Checks that the total number of 1 bits is even
 - If 2 errors occur at the same time it goes undetected
- Parity Blocks
 - Rectangle made up of rows
 - Last bit in each row is the parity bit
 - Parity byte is like parity bit but checks all the columns

5. Packet Switching and circuit switching

- When a message is sent over a WAN the message passes through multiple computers
- Packet switching
 - Splits the message into a number of equal sized packets
 - Each packet has its destination and packet sequence number
 - These packets are split and sent along communication lines
 - When packet reaches a node, node decides where to send it
 - The packets reassemble at the destination
 - Optimum use of connections around network
 - Maximises number of routes
 - More secure
 - Must take longer if 1 packet is slow
- Circuit switching
 - Creates a path for the message to be sent to the destination
 - Sends whole message
 - Fast
 - Insecure
 - Does not need reassembling

CHAPTER 1.6

1. Systems development life cycle
 - Waterfall model
 - Jackson Diagram
2. Problem Definition
 - Identify the problem or need for a new system.
 - Define objectives, scope, and constraints.
 - Example: A school wants an automated attendance system
3. Feasibility Study
 - Assess whether the proposed system is technically, economically, and operationally feasible.
 - Consider cost, time, resources, and benefits.
 - Example: Can the attendance system be developed within budget and time?
4. Requirement Analysis
 - Gather detailed functional (what the system should do) and non-functional (performance, security) requirements.
 - Techniques: Interviews, surveys, observation.
 - Example: The system must record attendance, generate reports, and notify absentees.
5. System Design
 - Plan the system architecture (e.g., databases, user interfaces, algorithms)
 - Use data flow diagrams (DFDs), entity-relationship (ER) diagrams, and pseudocode.
 - Example: Design a database schema for student records.
6. Implementation (Coding)
 - Write the actual code based on the design.
 - Follow programming best practices (modularity, documentation).
 - Example: Develop the attendance system using Python and SQL.
7. Testing
 - Verify that the system works correctly.
 - Types: Unit testing (individual components), integration testing (combined modules), system testing (full system), and user acceptance testing (UAT).
 - Example: Test if the attendance system correctly updates records.

8. Deployment

- Install the system in the real environment.
- Train users and migrate data if needed.
 - Example: Roll out the system in the school and train teachers.

9. Maintenance

- Fix bugs, improve performance, and add new features.
- Types: Corrective (bug fixes), adaptive (updates for new OS), perfective (enhancements).
 - Example: Add facial recognition to the attendance system.

CHAPTER 1.7

1. Custom Written Software
 - Expensive
 - Takes longer to become available
 - Bugs in software may not be identified for a long time
 - Very little to none help is available
2. Off-the-shelf software
 - Available immediately
 - Used by many people and organisations
 - Cheap
 - Bugs are rectified more quickly
 - Abundant amount of help on the internet
3. Business, Commercial and industrial applications
 - Stock control
 - Keep track of stock
 - Data about individual items
 - Data related to the product
 - Data about the use of the goods ie. when it was sold etc
 - Order Processing
 - Integrated closely to stock control
 - When the goods are delivered ordering system is informed so that the live order can be shown to have been fulfilled
 - Payroll
 - Employee records in 1 run because employees all need to be paid at the same time
 - Each record undergoes same type of processing
 - Data on number of hours, days, tax, etc
 - Processing requires no human interaction
 - Electronic point-of-sale (EPOS)
 - Identify goods being bought
 - Carry out process to produce satisfactory output
 - Receive customer's payment
 - Marketing
 - Produce advertising literature or images to put on the web
 - Computer Aided Design (CAD)
 - Use computer systems to design an item
 - Make calculations
 - Decisions about the way product is manufactured
 - Make predictions
 - Cost a solution (prices of individual components)
 - Computer Aided Manufacture (CAM)
 - Use computers to manufacture something

- Process Control
 - Control processes
 - Use sensors and make logical decisions
 - Continual cycle
 - Compares its previous actions and sees whether any impact was made
 - Precise measurements
 - Performs same repetitive task with no boredom or complaints
 - Immediately take actions when something is wrong
- 4. Generic Applications Software
 - Word Processor
 - Text editor
 - Microsoft word
 - Spreadsheet
 - Microsoft Spreadsheet
 - Desktop Publisher (DTP)
 - Adobe Illustrator
 - Able to manipulate images, text, sheets etc on a page(s)
 - Presentation Software
 - Google Slides
 - Powerpoint
 - Drawing Packages
 - Microsoft paint
 - Bitmap graphics
 - When enlarging image it makes the pixel sizes bigger
 - Vector graphics
 - When enlarging image no change
- 5. Generic Applications Software are not appropriate for specialised problems

CHAPTER 1.8

1. Automatic recording of data
 - Sensors
 - Sampling
 - Analogue to digital
 - Data logging, when sensors cant live stream its data, it stores it
 - Barcodes
 - Data can be inputted into the reader without any further preparation
 - Voice recognition
 - Uses a microphone to communicate with the computer
 - Sound waves are sampled and digitised and then compared with a library of digital values
 - Background sound can distort
 - Optical Mark Recognition (OMR)
 - Recognise dark spots/marks on a piece of paper
2. Manual data entry
 - Form based
3. Types of errors
 - Reasonable but wrong (human error)
 - Original data was inputted wrong
 - Johns bday was 18th Nov but was inputted as 8th Nov
 - Transcription Error
 - Human error
 - Can be solved by asking the operator to type it in twice
 - Nonsensical error
 - Johns bday is entered as 18/13/1998 theres only 12 months in a year so 13 is impossible
4. Data verification
 - Check the current inputted data with the original data
5. Data verification
 - Parity bits/bytes
 - Range check
 - A maths exam is scored from 0 to 100
 - A simple check can be used to check is a students score is within that range
 - If its not then there is an error
 - Character Check
 - If a field requires numbers only
 - A check can be made to make sure that there are only 0123456789
 - If somebody entered “o123” instead of “0123” we can check it

- Format Check
 - The field requires a date and time in DD/MM/YYYY
 - Some american put MM/DD/YYYY
- Length Check
 - Field requires 9-16 characters'
 - Somebody's password is only 8 characters long
- Existence Check
 - Barcode is scanned and is compared to a list
 - If list also has the code then barcode exists
- From a list Check
 - Drop down list in google sheets
- Presence Check
 - Checks whether something exists
 - Google forms have required fields
- Uniqueness Check
 - New product must have a unique ProductID

CHAPTER 1.9

1. Designing the user Interface

- The User Interface (UI) is the point of interaction between the user and the system
- A well-designed UI improves usability, accessibility, and user experience (UX)
- Poor UI design can lead to user frustration, errors, and inefficiency

2. Key Principles of UI Design

- Clarity
 - The interface should be easy to understand with clear labels and instructions.
 - Avoid unnecessary complexity.
- Consistency
 - Maintain uniformity in layout, colors, fonts, and navigation across the system.
 - Follow platform-specific design guidelines (e.g., Windows, macOS, mobile).
- Feedback
 - Provide immediate feedback to user actions (e.g., button clicks, error messages, loading indicators).
- Efficiency
 - Minimize the number of steps needed to complete a task.
 - Use shortcuts (e.g., keyboard shortcuts) for expert users.
- Accessibility
 - Ensure the UI is usable by people with disabilities (e.g., screen reader support, high-contrast modes).
- Error Prevention & Recovery
 - Design forms and inputs to prevent errors (e.g., input validation).
 - Provide clear error messages and ways to recover (e.g., undo function).

3. Types of User Interfaces

- Graphical User Interface (GUI)
- Command-Line Interface (CLI)
- Natural Language Interface (NLI)
- Menu-Driven Interface

4. UI Design Process

- Understand User Needs
 - Identify target users and their requirements
- Create Wireframes & Prototypes
 - Sketch low-fidelity wireframes (paper or digital) to plan layout.
 - Develop interactive prototypes (using tools like Figma, Adobe XD).
- Evaluate & Test
 - Conduct usability testing with real users.
 - Gather feedback and refine the design.

5. Common UI Elements

- Input Controls: Buttons, text fields, checkboxes, dropdown lists.
- Navigation Components: Menus, tabs, breadcrumbs.
- Informational Components: Tooltips, progress bars, notifications.
- Containers: Accordions, panels, modal windows.

6. Best Practices for UI Design

- Minimalism: Avoid clutter; focus on essential elements.
- Responsive Design: Ensure UI works on different screen sizes (desktop, tablet, mobile).
- Readable Typography: Use clear fonts and appropriate sizes.
- Color & Contrast: Ensure text is readable, and colors convey meaning (e.g., red for errors).

7. Evaluation of UI Design

- Effectiveness: Can users complete tasks successfully?
- Efficiency: How quickly can tasks be performed?
- Satisfaction: Do users find the interface pleasant to use?

Woohoo you can do it baabe!! Yayyyyyy i love u yipee!!

CHAPTER 1.10

1. Gates

- NOT
 - $1 = 0$
 - $0 = 1$
- AND
 - $0\ 0 = 0$
 - $1\ 0 = 0$
 - $0\ 1 = 0$
 - $1\ 1 = 1$
- OR
 - $0\ 0 = 0$
 - $1\ 0 = 1$
 - $0\ 1 = 1$
 - $1\ 1 = 1$
- NAND
 - $0\ 0 = 1$
 - $1\ 0 = 1$
 - $0\ 1 = 1$
 - $1\ 1 = 0$
- NOR
 - $0\ 0 = 1$
 - $1\ 0 = 0$
 - $0\ 1 = 0$
 - $1\ 1 = 0$
- XOR
 - $0\ 0 = 0$
 - $1\ 0 = 1$
 - $0\ 1 = 1$
 - $1\ 1 = 0$
- XNOR
 - $0\ 0 = 1$
 - $1\ 0 = 0$
 - $0\ 1 = 0$
 - $1\ 1 = 1$

2. Uses

- To calculate the outcome from a set of logic gates, make sure you calculate the intermediate processes
- Logic gates can be used within the processor as a form of refreshable memory and as an accumulator